

SOC Incident Response Report

Brute Force Authentication Attack

Incident Metadata

Incident ID	SOC-001
Incident Type	Brute Force Authentication Attack
Severity	Medium
Status	Closed
Date of Analysis	2026-01-23
Analyst	Prit Gujarati

Disclaimer

Environment & Scope Disclaimer This incident investigation was conducted as part of a controlled security practice exercise in a non-production lab environment. The target system (Kali Linux server) and attacker system were both owned and operated by the analyst for the purpose of testing detection, investigation, and incident response workflows. Security controls such as SSH access restrictions and firewall rules were intentionally configured to allow simulated attacker behavior. No production systems, real user data, or external networks were involved in this exercise. The findings and response actions documented in this report are based solely on log evidence generated during the controlled simulation.

1. Executive Summary

A brute-force SSH authentication attack was detected against a Linux server, where multiple failed login attempts were followed by a single successful authentication. Immediate containment actions were taken, including blocking the attacker IP via UFW, resetting the compromised account password, and locking the affected user account. Post-incident analysis confirmed no evidence of privilege escalation, command execution, file access, or data exfiltration.

2. Incident Overview

Field	Details
Attack Vector	Credential-based brute force
Target System	Kali Linux Server
Target Account	kali
Detection Method	Splunk correlation alert
Data Source	/var/log/auth.log
MITRE ATT&CK	T1110 – Brute Force

3. Attack Timeline (Log-Derived)

Based strictly on log timestamps

Time (UTC)	Event Description
03:32:51	Failed SSH password attempt
03:33:13	Multiple failed logins from same source IP
03:34:10	Successful SSH login (Accepted password)
03:34:30	SOC alert triggered

4. Detection & Evidence

Detection Logic

- ≥5 failed SSH login attempts
- Followed by ≥1 successful login
- From the same source IP and user
- Within a short time window

Key Log Evidence

- Failed password for kali from 10.89.146.71
- Accepted password for kali from 10.89.146.71

Splunk Fields Observed

src_ip	10.89.146.71
user	kali
sourcetype	linux_auth
source	/var/log/auth.log

5. Indicators of Compromise (IOCs)

IOC Type	Value
Source IP Address	10.89.146.71
Target Username	kali
Target Host	kali
Target Port	22 (SSH)
Log Source	auth.log

6. Technical Analysis

Analysis of authentication logs revealed a pattern consistent with an SSH brute force attack. The attacker attempted multiple password guesses in quick succession and eventually succeeded in authenticating using valid credentials. Post-authentication investigation included review of sudo logs, shell command history, and file activity logs. No evidence of privilege escalation, suspicious command execution, persistence mechanisms, or file deletion was found.

7. SOC Response Timeline

Reconstructed SOC response actions

Time (UTC)	Action Taken
03:34	Splunk alert triggered
03:35	SOC analyst initiated investigation
03:36	Source IP identified
03:36	IP blocked using UFW
03:37	User password changed
03:38	User account locked (passwd -L kali)
03:40	Full log review completed
03:45	Incident closed

8. Impact Assessment

Successful authentication occurred
No malicious commands executed
No file deletion or modification
No data exfiltration
No persistence mechanisms created

9. Mitigation Actions

Blocked attacker IP using UFW firewall
Changed password of affected account
Locked compromised user account
Reviewed SSH authentication configuration
Verified system integrity via log analysis

10. Recommendations

Disable password-based SSH authentication
Enforce SSH key-based authentication
Implement Fail2Ban for automated blocking
Reduce allowed SSH authentication attempts

11. MITRE ATT&CK Mapping

Tactic	Technique ID	Technique Name
Credential Access	T1110	Brute Force
Persistence (Attempted)	T1078	Valid Accounts

12. Conclusion

This incident demonstrates effective SOC monitoring and response capabilities. Although the attacker achieved SSH authentication, immediate detection and containment prevented any further compromise. Continuous monitoring, strong authentication controls, and rapid response remain critical to mitigating credential-based attacks.

Appendix

Log Source: /var/log/auth.log
Detection Type: SIEM Correlation Alert
Analysis Type: Post-Incident Log Analysis